

SEMIYARKA, Kazakhstan

WMO: 361520

Lat: 50.8927N

Lon: 78.3222E

Elev: 489

StdP: 14.44

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.9	(c) -26.5	(d) -38.9	(e) 0.6	(f) -32.8	(g) -32.5	(h) 0.9	(i) -26.2	(j) 26.9	(k) 16.8	(l) 23.1	(m) 18.3	(n) 4.0	(o) 140	(p) 0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.2	(c) 93.4	(d) 64.8	(e) 90.0	(f) 64.0	(g) 86.7	(h) 63.1	(i) 68.0	(j) 83.4	(k) 66.7	(l) 82.4	(m) 65.5	(n) 80.9	(o) 8.8	(p) 50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 63.7	(b) 90.0	(c) 71.6	(d) 61.9	(e) 84.3	(f) 71.0	(g) 60.0	(h) 78.6	(i) 70.9	(j) 32.6	(k) 82.7	(l) 31.6	(m) 82.3	(n) 30.6	(o) 81.1	(p) 94.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.7	18.6	-34.4	99.5	9.0	3.0	-40.9	101.7	-46.1	103.4	-51.1	105.1	-57.7	107.3
			-34.6	71.2	8.9	2.6	-41.0	73.1	-46.2	74.6	-51.2	76.1	-57.6	78.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.8	1.6	4.3	22.7	45.7	58.1	69.7	71.8	68.3	56.3	42.4	25.0	9.9		
	DBStd	27.38	16.10	15.24	13.57	10.36	8.50	7.16	6.02	6.83	8.53	9.25	13.41	14.07		
	HDD50	6149	1500	1281	848	197	26	0	0	0	36	267	751	1243		
	HDD65	9789	1965	1701	1313	581	244	38	12	43	282	701	1201	1708		
	CDD50	2433	0	0	0	68	275	590	676	568	224	32	0	0		
	CDD65	598	0	0	0	2	29	178	223	146	20	0	0	0		
	CDH74	7177	0	0	0	61	519	2077	2367	1770	374	9	0	0		
	CDH80	2867	0	0	0	13	157	855	987	727	127	1	0	0		
	WSAvg	8.4	7.8	8.2	8.8	9.2	9.2	8.5	8.1	7.9	8.0	8.2	8.6	8.7		
	PrecAvg	8.9	0.4	0.4	0.4	0.5	0.9	1.1	1.4	1.1	0.8	0.8	0.7	0.5		
	PrecMax	14.9	1.4	1.1	1.7	1.6	3.0	3.0	3.4	3.6	1.9	1.5	1.4	1.3		
(15) Precipitation	PrecMin	5.6	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.2	0.0	0.0		
	PrecStd	1.9	0.3	0.3	0.3	0.4	0.7	0.9	0.9	0.9	0.5	0.4	0.3	0.3		
	DB	36.1	37.4	60.6	80.9	91.7	96.1	97.1	97.5	90.5	74.1	55.8	37.8			
	MCWB	33.1	34.1	48.3	56.7	62.2	64.4	66.8	65.9	61.6	53.3	45.3	34.6			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	32.9	34.0	51.3	73.9	84.5	92.0	92.7	92.3	83.3	67.0	50.3	35.0			
	2% MCWB	30.4	31.1	42.1	54.3	59.7	63.5	65.7	64.8	58.5	50.7	43.4	32.5			
	5% DB	28.7	30.5	43.6	68.5	80.1	88.4	89.2	87.7	77.8	61.9	44.8	32.1			
	5% MCWB	26.3	28.0	38.6	51.4	58.1	63.1	64.9	63.4	56.7	48.6	39.7	29.7			
	10% DB	24.3	26.7	38.8	63.8	75.8	84.9	85.6	83.7	72.5	56.7	40.6	29.4			
	10% MCWB	22.2	24.8	35.2	49.4	56.6	62.6	64.1	62.2	54.4	46.1	36.6	27.3			
	0.4% WB	33.7	34.6	48.7	59.1	64.3	69.1	70.6	68.7	64.1	55.6	48.7	35.3			
	0.4% MCDB	35.8	36.8	59.5	75.5	81.3	84.5	86.4	88.3	84.8	68.7	52.9	37.4			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	30.5	31.0	42.9	55.6	61.8	66.9	68.4	66.7	60.3	52.2	43.5	32.7			
	2% MCDB	32.9	33.5	49.4	71.0	80.5	83.8	82.5	84.9	78.1	64.9	49.5	34.8			
	5% WB	26.3	28.1	38.6	52.8	59.8	65.5	67.3	65.3	58.0	49.3	40.0	29.9			
	5% MCDB	28.6	30.4	43.2	66.3	75.6	82.0	81.5	82.3	74.4	59.9	44.8	31.9			
	10% WB	22.3	24.7	35.4	50.1	57.8	64.1	66.1	63.8	55.9	46.8	36.7	27.1			
	10% MCDB	24.2	26.8	38.7	62.1	72.6	79.6	80.7	80.1	70.6	56.1	40.3	29.4			
	MDBR	17.3	19.2	17.7	23.2	24.5	22.6	21.2	23.4	23.2	18.7	14.4	16.6			
	5% DB	16.9	18.0	20.8	31.6	31.7	28.9	27.3	30.3	32.4	28.1	18.7	15.5			
(36) Mean Daily Temperature Range	MCWBR	15.4	16.1	13.7	15.5	12.9	9.6	8.2	10.2	13.6	14.6	13.2	14.5			
	MCDBR	17.4	17.6	20.2	29.0	27.3	23.7	22.0	25.6	28.3	25.0	18.1	16.2			
	MCWBR	16.0	16.0	13.8	15.1	13.0	9.2	8.2	9.7	13.2	14.0	13.1	15.4			
	MCDBR	16.0	16.0	13.8	15.1	13.0	9.2	8.2	9.7	13.2	14.0	13.1	15.4			
(40) Clear-Sky Solar Irradiance	taub	0.304	0.343	0.399	0.423	0.428	0.427	0.452	0.423	0.377	0.357	0.344	0.315			
	taud	2.054	1.965	1.977	2.099	2.137	2.222	2.161	2.216	2.289	2.247	2.131	2.044			
	Ebn at Noon	214	236	246	258	263	264	254	255	255	234	197	186			
	Edh at Noon	26	38	46	46	46	43	45	40	33	29	25	23			
(44) All-Sky Solar Radiation	RadAvg	437	796	1213	1533	1946	2059	1877	1682	1254	709	373	303			
	RadStd	39	57	96	111	135	116	131	78	99	88	40	37			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page